Explanation:

- The function chooses the middle element as the pivot and recursively sorts the left and right partitions of the list.

---

## Step 3: Searching Algorithms

Searching algorithms are used to find specific elements in a dataset. We will implement two common searching algorithms.

### 1. Linear Search

Linear Search checks each element of the list one by one until it finds the target value.

```
def linear_search(data, target):
    for i, value in enumerate(data):
        if value == target:
                return i  # Return index of the target element
    return -1  # Return -1 if the target is not found
```

Explanation:

- The function loops through each element and returns the index if the target is found. If the target is not found, it returns -1.